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Introduction

Our project re-implements the core method from *DoRA: Weight-Decomposed Low-Rank Adaptation* by Shih-Yang Liu, Chien-Yi Wang, Hongxu Yin, Pavlo Molchanov, Yu-Chiang Frank Wang, Kwang-Ting Cheng, and Min-Hung Chen. The paper addresses a limitation of LoRA, which makes fine-tuning cheaper by training low-rank adapter matrices instead of updating the full model, but can be limited by its single low-rank update branch. DoRA improves on LoRA by decomposing pretrained weights into a learned magnitude component and a direction component updated with LoRA-style low-rank adaptation. This increases flexibility while preserving parameter-efficient fine-tuning. Our reproduction focuses on a smaller-scale commonsense-reasoning setting, where we implement DoRA, compare it against LoRA, and evaluate performance on multiple-choice reasoning tasks.

Chosen Result

We chose to reproduce the commonsense-reasoning comparison between DoRA and LoRA from Table 1 of the original paper. This table reports accuracy on eight multiple-choice reasoning datasets: BoolQ, PIQA, SIQA, HellaSwag, WinoGrande, ARC-Easy, ARC-Challenge, and OpenBookQA. The key result is that DoRA improves over LoRA on average while using a similar number of trainable parameters. For example, on LLaMA-7B, LoRA achieves 74.7 average accuracy, while DoRA achieves 78.4.

Table 1. Accuracy comparison of LLaMA 7B/13B, LLaMA2 7B, and LLaMA3 8B with various PEFT methods on eight commonsense reasoning datasets. Results of all the baseline methods on LLaMA 7B/13B are taken from (Hu et al., 2023). Results of LoRA on LLaMA2 7B and LLaMA3 8B are obtained using the hyperparameters described in (Hu et al., 2023). DoRA†: the adjusted version of DoRA with the rank halved.

Model	PEFT Method	# Params (%)	BoolQ	PIQA	SIQA	HellaSwag	WinoGrande	ARC-e	ARC-c	OBQA	Avg.
ChatGPT	-	-	73.1	85.4	68.5	78.5	66.1	89.8	79.9	74.8	77.0
LLaMA-7B	Prefix	0.11	64.3	76.8	73.9	42.1	72.1	72.9	54.0	60.6	64.6
	Series	0.99	63.0	79.2	76.3	67.9	75.7	74.5	57.1	72.4	70.8
	Parallel	3.54	67.9	76.4	78.8	69.8	78.9	73.7	57.3	75.2	72.2
	LoRA	0.83	68.9	80.7	77.4	78.1	78.8	77.8	61.3	74.8	74.7
	DoRA† (Ours)	0.43	70.0	82.6	79.7	83.2	80.6	80.6	65.4	77.6	**77.5**
	DoRA (Ours)	0.84	69.7	83.4	78.6	87.2	81.0	81.9	66.2	79.2	**78.4**

Table 1. *Cropped Table 1 from the DoRA paper. The LLaMA-7B rows show DoRA improving over LoRA in average accuracy, motivating our scaled-down DoRA vs. LoRA reproduction.*

We selected this result because it directly tests the paper's central claim: separating weight magnitude from weight direction makes low-rank adaptation more expressive than standard LoRA. In DoRA, the adapted weight can be written as:

$$W' = m * (W_0 + \Delta V) / ||W_0 + \Delta V||$$

Here, m is the learned magnitude term, W_0 is the frozen pretrained weight, and ΔV is the LoRA-style low-rank directional update. Our implementation uses a smaller course-project-scale setup, but the evaluation question is the same: does DoRA outperform LoRA when both methods are trained under comparable conditions?

Methodology:

To reproduce the results found in the paper, we implemented both DoRA and LoRA using the same hyperparameters and compared our accuracy on these reasoning tasks with the results reported in the paper. We first experimented with the RoBERTa roberta-base model before transitioning to the Microsoft Phi-2 model for our main experiments in order to reduce training time and compute cost compared to the Llama model used in the paper.

We follow DoRA's decomposition of each linear weight into a learnable magnitude vector and a directional component built from the frozen pretrained matrix W0. The magnitude is initialized from the L2 row norms of W0, while adaptation enters through a low-rank update BA applied to the direction. After adding that update, the direction is renormalized row-wise so that only the magnitude scales the effective weight, preserving the intended geometry of the update. In both LoRA and DoRA, we initialized the B matrix to all zeros while initializing the A matrix with Kaiming uniform initialization to avoid near-zero gradients and allow the model to learn effectively at the beginning of training.

The remainder of the PEFT parameters were taken directly from the paper to give our implementation the best chance of replicating the results. We chose 16 as the rank of these A/B matrices and 32 as our alpha, and trained the model for 3 epochs, consistent with the training settings used in the paper. Finally, we set the learning rate to 2e-4 with a linear learning rate scheduler that decays to 0 following a warmup period. We used the AdamW optimizer, along with gradient checkpointing and 4-bit loading for the Microsoft Phi-2 model to reduce memory usage and enable training under limited compute resources.

We trained three models total: our DoRA implementation fine-tuned on Phi-2 (2.7B), and both our LoRA and DoRA implementations fine-tuned on RoBERTa-base (125M). All models are trained on the combined 8 datasets using the same hyperparameters to ensure a fair comparison. We then evaluate each model on all 8 tasks individually.

Results & Analysis:

We reproduced the commonsense reasoning experiments from the DoRA paper using the Phi-2 language model instead of the paper's LLaMA-7B backbone. As mentioned, to make the comparison as direct as possible, we kept the main training parameters the same as the paper, including the rank, alpha, and learning rate. Using Phi-2 with DoRA adapters, we obtained the results shown in Table 2. Across the eight commonsense reasoning datasets, our implementation achieved a higher average score than the paper's reported LLaMA-7B DoRA results.

Method	BoolQ	PIQA	SIQA	HellaSwag	WinoGrande	ARC-Challenge	ARC-Easy	OpenBookQA	Avg.
Paper DoRA (LLaMA-7B)	70.0%	82.6%	79.7%	83.2%	80.6%	65.4%	80.6%	77.6%	**77.5%**
Our DoRA (Phi-2)	82.3%	84.5%	75.6%	88.6%	80.0%	75.5%	86.0%	70.2%	**80.3%**

Table 2. *Comparison between the paper's reported LLaMA-7B DoRA results and our reproduced Phi-2 DoRA implementation across eight commonsense reasoning benchmarks.*

Due to compute limitations and long training times on the A100 GPU available through Google Colab, we were only able to run the full Phi-2 training setup once.

Before switching to Phi-2, we initially implemented both DoRA and our own custom LoRA implementation on the smaller roberta-base model. These results were significantly worse than those reported in the paper, largely because roberta-base is a much smaller encoder-only model, while the paper evaluates larger decoder-style language models that are better suited for commonsense reasoning tasks. The results in Table 3 show that although overall performance was lower than the larger language models

used in the paper, DoRA still consistently outperformed both the base model and our LoRA implementation across most datasets.

Method	BoolQ	PIQA	SIQA	HellaSwag	WinoGrande	ARC-Challenge	ARC-Easy	OpenBookQA	Avg.
Base	49.9%	51.7%	32.1%	24.3%	50.5%	26.1%	27.2%	23.0%	**35.6%**
LoRA	69.8%	66.7%	70.8%	59.2%	61.0%	32.1%	46.8%	51.6%	**57.3%**
DoRA	71.1%	67.2%	70.9%	60.1%	60.7%	36.3%	46.8%	50.1%	**57.9%**

Table 3. *Results of our RoBERTa-base experiments comparing the base model, LoRA, and DoRA on the same commonsense reasoning benchmarks.*

The main challenges during reproduction were first understanding why roberta-base was underperforming, and then dealing with computational limitations after transitioning to Phi-2. Despite these constraints, our final results still support the paper's main contribution: separating weight magnitude and direction improves the quality of low-rank adaptation over standard LoRA. While our Phi-2 setup slightly underperformed some newer, larger models in the paper, it outperformed the reported LLaMA-7B DoRA results and was competitive with newer models like LLaMA2-7B and LLaMA3-8B.

Reflections:
Looking back, the biggest lesson from this project was how much the choice of base model matters when reproducing PEFT results. We initially assumed that as long as we implemented DoRA correctly, the method itself would carry the results. The roberta-base experiments quickly showed that this wasn't true. Even though DoRA still beat LoRA on that smaller encoder model, the absolute scores were far below the paper's numbers, which made it hard to tell whether our implementation was actually working as intended. Switching to Phi-2 made the comparison feel much more meaningful and lined up more closely with the spirit of the original paper.

The other major lesson was around compute. Running large language model fine-tuning on a single A100 through Colab was a real bottleneck. We were only able to do one full Phi-2 training run, which means our final numbers should be read as a single sample rather than a stable average. If we had more compute, we would have run multiple seeds for both LoRA and DoRA to get a cleaner head to head comparison.

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